A Curvature Decomposition of the Explicit Formula

for the Riemann Zeta Function

Ulrich Tehrani

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Abstract

We study the second logarithmic derivative of the completed Riemann zeta function in the form

$$H_\xi(\sigma,t) := \partial_\sigma^2 \log|\xi(\sigma+it)|^2,$$

and interpret it as a second-derivative specialization of the classical explicit formula. This yields a prime-cutoff decomposition into local curvature contributions from the archimedean place and finite Euler factors, together with a spectral remainder encoding the contribution of the nontrivial zeros. We prove local positivity at every finite place, establish critical divergence of the truncated local curvature on the line $\sigma = \frac{1}{2}$ with order $(\log\kappa)^2$, and show convergence for $\sigma > \frac{1}{2}$. We also discuss the automorphic Rankin–Selberg analogue and the structural proximity of the curvature field to Li-type positivity criteria. The note concludes with an open question on whether the singular kernel underlying this curvature formulation can be embedded into the admissible Weil test-function framework.

1 The Curvature Field

The starting point is the second logarithmic derivative of ξ, which by the Hadamard product decomposes as

$$(\log\xi)''(s) = -\sum_\rho \frac{1}{(s-\rho)^2} + (\text{local terms}). \tag{0}$$

Equation (0) shows that the curvature field H_ξ defined below can be viewed as the second-derivative specialization of the classical explicit formula. The zero sum encodes the nontrivial zero geometry, while the remaining local terms arise from the archimedean contribution and the local Euler factors. The present note studies the real part of this identity as a curvature field.

The Hadamard product representation together with logarithmic differentiation of the completed zeta function $\xi(s)$ yields

$$H_\xi(\sigma,t) := \frac{d^2}{d\sigma^2}\log|\xi(\sigma+it)|^2 = 2\,\Re(\log\xi)''(s) = -2\,\Re\sum_\rho \frac{1}{(s-\rho)^2} + (\text{archimedean and trivial-zero terms}), \tag{1}$$

where the sum runs over the nontrivial zeros ρ of $\zeta(s)$. The archimedean contribution (from the Gamma factor) is separated into the local terms defined below. This quantity may be viewed as a curvature-type specialization of the spectral side of the explicit formula: choosing a test function whose Mellin transform produces the kernel $(s-\rho)^{-2}$ recovers H_ξ as the spectral output.

In the adelic framework of Tate's thesis, define local curvature contributions at each place. At the archimedean place,

$$\psi_1(\sigma) := \frac{1}{4}\,\frac{d^2}{d\sigma^2}\log|\Gamma(\sigma/2)|^2.$$

For real σ this coincides with $\frac{1}{8}\psi^{(1)}(\sigma/2)$, where $\psi^{(1)}$ denotes the trigamma function. (One has $\frac{d^2}{d\sigma^2}\log|\Gamma(\sigma/2)|^2 = \frac{1}{2}\psi^{(1)}(\sigma/2)$ by the chain rule, so $\psi_1(\sigma) = \frac{1}{4}\cdot\frac{1}{2}\psi^{(1)}(\sigma/2) = \frac{1}{8}\psi^{(1)}(\sigma/2)$.)

At each finite place p, with M_s the local Mellin transform and f_p the standard Tate test function,

$$V_p(\sigma) := \frac{d^2}{d\sigma^2}\log\|M_s f_p\|^2_{L^2(\mathbb{Q}_p^\times)}. \tag{2}$$

Explicitly, for the unramified local factor,

$$\|M_s f_p\|^2 = (1-p^{-1})(1-p^{-2\sigma})^{-1}.$$

so that

$$V_p(\sigma) = 4(\log p)^2\,\frac{p^{-2\sigma}}{(1-p^{-2\sigma})^2}. \tag{3}$$

Define the truncated local curvature

$$H_{\text{local}}(\sigma,\kappa) := \psi_1(\sigma) + \sum_{p\le\kappa} V_p(\sigma) \tag{4}$$

and the spectral remainder at cutoff κ

$$H_{\text{dual}}(\sigma,t,\kappa) := H_\xi(\sigma,t) - H_{\text{local}}(\sigma,\kappa). \tag{5}$$

Note that for $\sigma > \frac{1}{2}$, $H_{\text{local}}(\sigma,\kappa)$ converges as $\kappa\to\infty$ and $H_{\text{dual}}(\sigma,t,\kappa)$ has a well-defined limit. At $\sigma = \frac{1}{2}$, H_{local} diverges (Lemma 2 below), and H_{dual} must be understood at finite κ; the decomposition $H_\xi = H_{\text{local}} + H_{\text{dual}}$ holds at every finite cutoff.

The prime-cutoff decomposition $H_\xi = H_{\text{local}} + H_{\text{dual}}$ separates the curvature into local prime contributions and the spectral remainder encoding the zero contributions. This is the corresponding specialization of the local side of the explicit formula to the same curvature-type test function. The decomposition holds at every finite cutoff κ and is not a new identity but a reorganization of the explicit formula in curvature form.

To the best of our knowledge, this curvature formulation does not appear explicitly in the literature (cf. J.-F. Burnol, personal communication).

2 Local Positivity

Lemma. *For every prime p and all $\sigma > 0$,*

$$V_p(\sigma) \geq 0.$$

Proof. From the explicit expression (3),

$$V_p(\sigma) = 4(\log p)^2 \frac{p^{-2\sigma}}{(1-p^{-2\sigma})^2}.$$

All factors are manifestly positive for $\sigma > 0$, since $0 < p^{-2\sigma} < 1$. More generally, for the active local norm with prime cutoff K and annulus values $S_m = p^{1-m} - p^{-K}$ (for $m = 1, \ldots, K$) and $S_0 = 2 - p^{-1} - p^{-K}$,

$$\|M_s f_{\kappa,p}\|^2 = (1-p^{-1})\Big(\frac{|S_0|^2}{1-p^{-2\sigma}} + \sum_{m=1}^{K} |S_m|^2\, p^{2m\sigma}\Big).$$

This is a sum of exponentials in σ with positive coefficients. Its logarithm is therefore convex: the second derivative of a logarithm of a sum of positive exponentials is non-negative by the Cauchy–Schwarz inequality applied to the variance of the exponential weights. $\square$

The archimedean contribution $\psi_1(\sigma) \geq 0$ is classical (log-convexity of the Gamma function, Bohr–Mollerup theorem).

Corollary. *For all $\sigma > 0$ and all κ,*

$$H_{\text{local}}(\sigma, \kappa) \geq 0.$$

3 Divergence and Convergence

Lemma. *As $\kappa \to \infty$,*

$$H_{\text{local}}(1/2, \kappa) \asymp C\,(\log \kappa)^2$$

for a constant $C > 0$ depending on the normalization.

Proof sketch. From (3),

$$V_p(1/2) \sim 4(\log p)^2\, p^{-1}(1-p^{-1})^{-2}$$

for large p. By the prime number theorem,

$$\sum_{p \leq \kappa} \frac{(\log p)^2}{p} \sim (\log \kappa)^2,$$

by partial summation against $\pi(x) \sim x/\log x$. The leading constant depends on the precise local norm conventions; numerically we observe $C \approx 2$ for the standard Tate test functions (with V_p as in (3)), but we state only the order of growth as proven.

Lemma. *For $\sigma > \frac{1}{2}$,*

$$H_{\mathrm{local}}(\sigma, \kappa) \to H_{\mathrm{local}}(\sigma, \infty) < \infty.$$

Proof. One has $V_p(\sigma) = O((\log p)^2 p^{-2\sigma})$. Since $2\sigma > 1$, the series $\sum_p (\log p)^2 p^{-2\sigma}$ converges absolutely by comparison with the prime zeta function. $\square$

Summary. The local curvature diverges at the critical point $\sigma = \frac{1}{2}$ and converges everywhere else in the half-plane $\sigma > \frac{1}{2}$. The critical line is the phase boundary between divergence and convergence of the local curvature sum.

4 Numerical Illustration

At $t = 0$ (away from zeros), $\sigma = 0.3$, $\kappa = 1300$,

$$H_{\mathrm{local}} \approx 915, \qquad H_\xi \approx 0.09.$$

The spectral remainder H_{dual} nearly cancels H_{local}, with compensation exceeding 99.99%.

Near the first zero $t = \gamma_1 \approx 14.13$,

$$H_\xi(0.3, \gamma_1) \approx -50, \qquad H_\xi(0.5, \gamma_1) \approx +89558.$$

The curvature spikes sharply at $\sigma = \frac{1}{2}$ and is negative on both sides. The spike arises because the term $2\Re(s-\rho)^{-2}$ in (1) has a double pole at $s = \rho$; near a zero, the dominant contribution is governed by the singular kernel $2\Re(s-\rho)^{-2}$, producing the large values observed. The spikes become sharper near higher zeros: $H_\xi(0.5, \gamma_2) \approx +480178$.

The symmetry $H_\xi(\sigma, t) = H_\xi(1-\sigma, t)$ is verified numerically to machine precision, as it must hold by the functional equation $\xi(s) = \xi(1-s)$.

5 Rankin–Selberg Universality

The divergence at $\sigma = \frac{1}{2}$ is not specific to $\zeta(s)$.

For a cuspidal automorphic representation π on $\mathrm{GL}(n)/\mathbb{Q}$ with Satake parameters $\alpha_{\pi,j}(p)$, the diagonal curvature sum

$$\sum_{p \le \kappa} \Big(\sum_{j=1}^{n} |\alpha_{\pi,j}(p)|^2 \Big) \frac{(\log p)^2}{p}$$

grows on the order of $C_\pi (\log \kappa)^2$ with $C_\pi > 0$. This follows from the simple pole of $L(s, \pi \times \tilde{\pi})$ at $s = 1$, combined with a Tauberian argument; the precise asymptotic coefficient depends on normalization conventions (cf. Kowalski–Michel, *Pacific J. Math.* 207(2), 2002, for related prime sums in the Rankin–Selberg context).

Thus the curvature decomposition and the critical divergence are universal phenomena for automorphic L-functions with Euler product and functional equation.

6 Connection to Li Coefficients

The Li coefficients λ_n, defined via

$$g(z) := \log \xi \left(\frac{1}{1-z} \right), \qquad g'(z) = \sum_{n \geq 1} \lambda_n z^{n-1},$$

satisfy Li's criterion:

$$\mathrm{RH} \iff \lambda_n \geq 0 \quad \text{for all } n \geq 1.$$

The second derivative yields

$$g''(z) = (\log \xi)'' \left(\frac{1}{1-z} \right) (1-z)^{-4} + 2(\log \xi)' \left(\frac{1}{1-z} \right) (1-z)^{-3}. \tag{6}$$

The term $(\log \xi)''(s)$ appearing here is exactly the core of the curvature field:

$$H_\xi(s) = 2\Re(\log \xi)''(s).$$

The present curvature quantities are therefore structurally close to Li-type positivity criteria: both are built from logarithmic derivatives of ξ. The difference is that $H_\xi(s)$ is a continuous pointwise second-derivative field, whereas Li's coefficients package the same logarithmic data into a discrete sequence after Mobius reparametrization around $s = 1$.

Important caveat. The local positivity $V_p(\sigma) \geq 0$ at each place does *not* directly imply $\lambda_n \geq 0$. Li positivity is a global statement mixing all places, the functional equation, the complete zero geometry, and specific test-function structure. Local positivity is a necessary ingredient but not sufficient. This gap is fundamental and is likely equivalent to RH itself.

7 The Open Question

The central open problem arising from this decomposition is the following.

Question. Does there exist an appropriate global packaging — a summation procedure, test-function class, or regularization — that transfers the local positivity

$$V_p(\sigma) \geq 0 \quad (\forall p), \qquad \psi_1(\sigma) \geq 0$$

and the critical divergence

$$H_{\mathrm{local}}(1/2, \kappa) \to \infty$$

into a global Li/Weil-type positivity statement?

More precisely: can the singular test function implicit in H_ξ — whose Mellin side produces the kernel $(s-\rho)^{-2}$ — be embedded in, or approximated by, the class of admissible test functions for which Weil positivity $W(f * \check{f}) \geq 0$ is equivalent to RH?

This is a question about the interaction between

- local curvature positivity (proven, Lemma 1),
- critical divergence rate (proven above),
- the functional equation, and
- the admissible test-function class.

A positive answer would connect the curvature decomposition to the Weil positivity framework (cf. Lagarias, 1999; Connes–Consani, 2021) and potentially yield RH.

References

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Note added. A companion note [8] constructs an explicit admissible test function $g^*_{\sigma,\varepsilon} \in S_{ad}$ and derives the prime-side identity $W(g^* * \check{g}^*) = Z(g^*) - H_{\text{local}}(\sigma, \kappa) + O(\varepsilon)$, connecting the curvature decomposition of the present note to the Weil positivity framework.

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